

Department of Electronics and Communication Engineering

QUESTION BANK

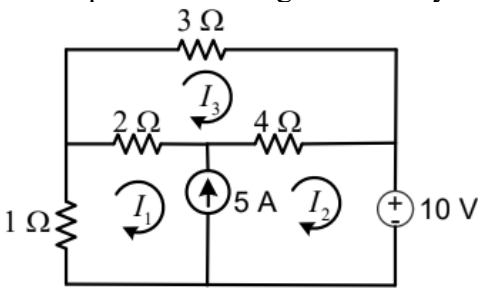
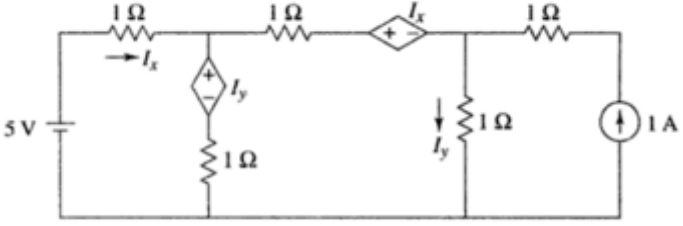
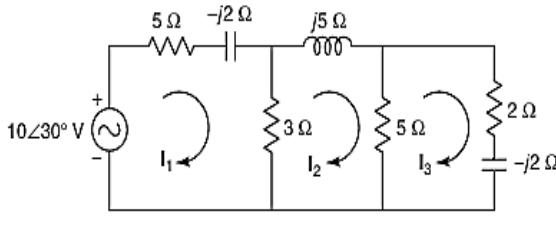
Branch	Semester	Course Name	Course Code
ECE	4	NETWORK ANALYSIS	BEC24402

Course Outcomes

CO1	Determine the current and/or voltage by simplifying an electrical network using mesh and node analysis.
CO2	Solve the electrical networks by applying theorems to reduce circuit complexities.
CO3	Determine Z, Y, h and T parameters and their inter relationship for a given two port networks.
CO4	Analyze the initial behaviour of the electrical circuit and find the network solution using Laplace transform
CO5	Estimate the current and/or voltage for the given electrical networks using Graph theory.

Module No. 1: Basic circuit analysis concepts

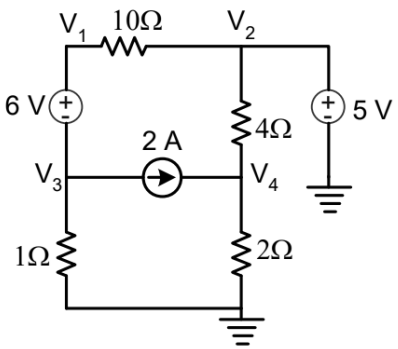
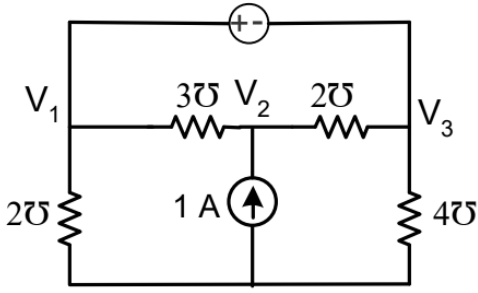
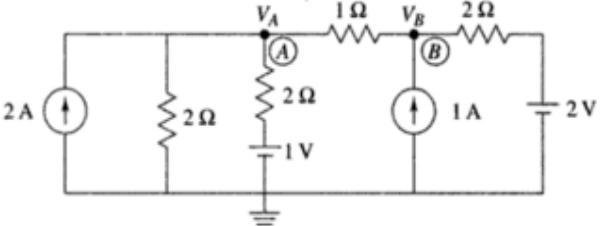
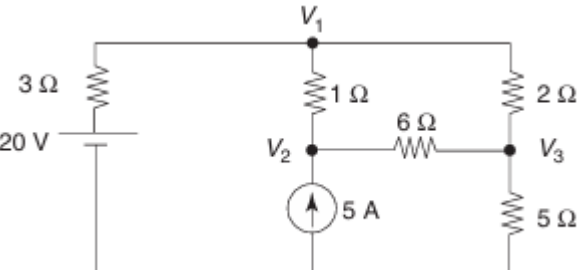
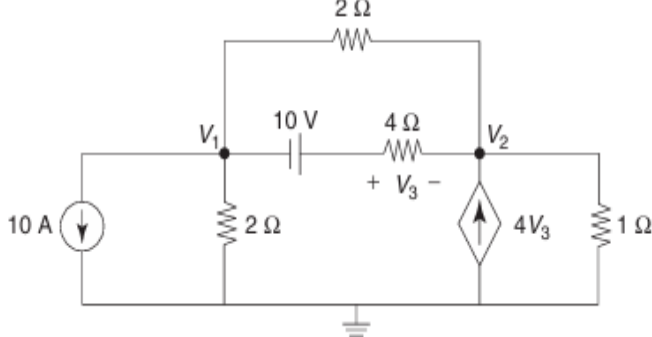
Unit-1:

Question No.	Questions	Marks	Course Outcome	RBT Level
1	Determine all the loop currents using Mesh analysis 	7	CO1	L3
2	Find the current in the three meshes for the figure Shown 	7	CO1	L2
3	Explain loop method of analysis with an example and solve 	7	CO1	L2
4	Determine the Loop currents	7	CO1	L2

5	Determine the Loop currents	7	CO1	L2
6	Determine the Loop currents and find the current through 2-ohm resistor	7	CO1	L2
7	Determine the Loop currents using loop method of analysis	7	CO2	L3
8	Explain the classification of electrical network	8	CO1	L2
9	Explain with an example Dependent, independent sources and super mesh.	8	CO1	L2

Unit-2

Question No.	Questions	Marks	Course Outcome	RBT Level
1	Apply Nodal Analysis to find Node Voltages V1, V2, And V3.	7	CO1	L2

2	For the circuit shown in Figure determine all the node voltages 	7	CO1	L2
3	Determine all the node voltages using nodal analysis 	7	CO1	L2
4	Find V_A and V_B using Nodal method 	7	CO1	L2
5	Determine the node voltages using Nodal method of analysis 	7	CO1	L2
6	 Determine the node voltages using Nodal method of analysis	7	CO1	L2
7	Determine the node voltage V_x using Nodal method of analysis	7	CO1	L2

8	Determine the node voltages using Nodal method of analysis.	7	CO1	L3

Unit-3

Question No.	Questions	Marks	Course Outcome	RBT Level
1	Find V_o using source Transformation	6	CO1	L2
2	Determine the Input resistance between PQ using star-delta transformation.	6	CO1	L2
3	For the circuit shown in Figure determine the equivalent resistance between any two terminals	6	CO1	L2
4	Determine the equivalent resistance between terminals A and B	6	CO1	L2

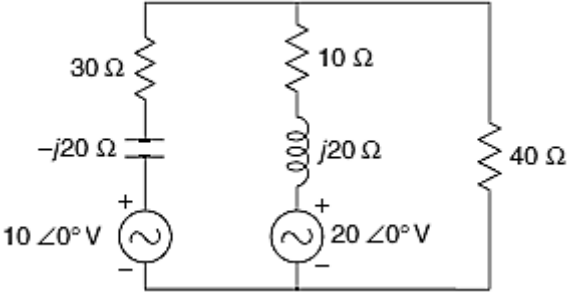
5	Find the resistance of the circuit shown in Figure between A and B	6	CO1	L2
6	Obtain the expression for star to delta conversion	6	CO1	L2
7	Determine the Power delivered by 50V source using Source transformation	6	CO1	L2
8	Reduce the network shown in to a single source network	6	CO1	L2

MODULE-2: Network Theorems

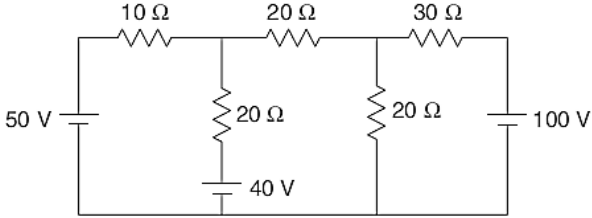
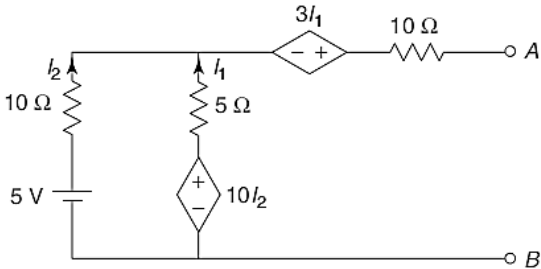
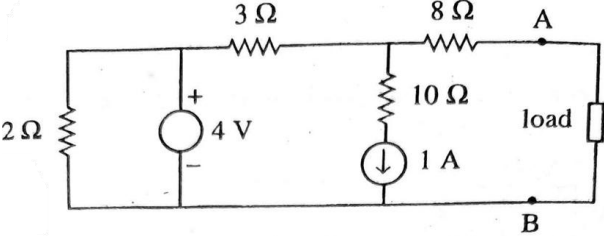
Unit-1

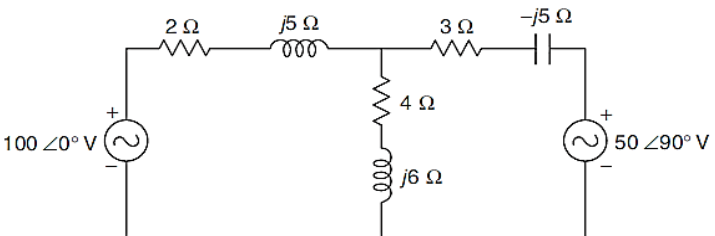
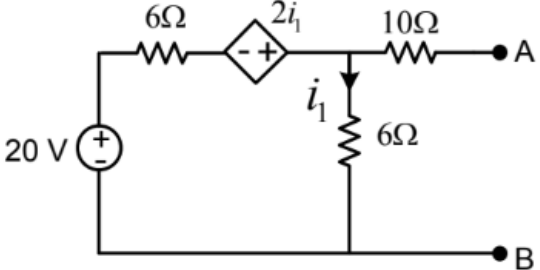
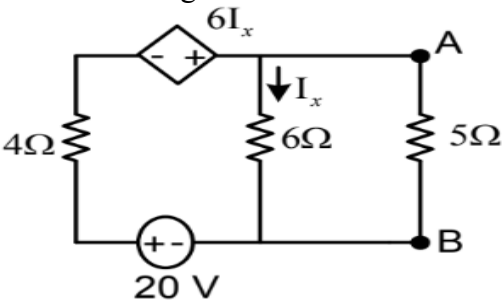
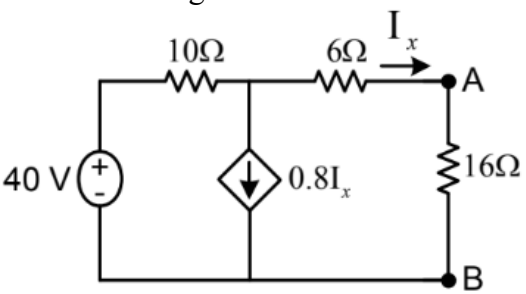
Question No.	Questions	Marks	Course Outcome	RBT Level
1	Find the current I and verify reciprocity theorem for the network shown	7	CO2	L3

2	Find the current through the 10Ω resistor in the network using Millman's theorem	7	CO2	L3
3	Verify Reciprocity theorem by finding the current through the branch b c in the circuit shown.	7	CO2	L3
4	Find the output voltage E_0 for the circuit shown using Millman's Theorem.	7	CO2	L3
5	Using Millman's theorem find the current through the load resistance $R_L=100\Omega$ in the circuit	7	CO2	L2
6	State and prove Reciprocity theorem.	7	CO2	L2
7	Using Millman's theorem find current through R_L for the circuit shown in Figure.	7	CO2	L3

8	Find the current through the 40Ω resistor for the network shown in Fig. using Millman's theorem. 	8	CO2	L3

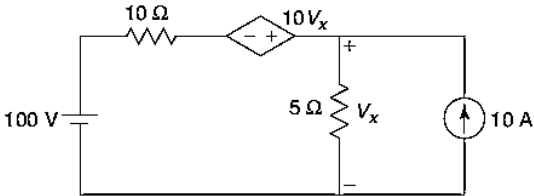
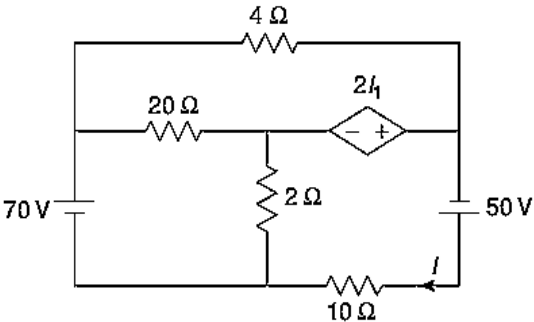
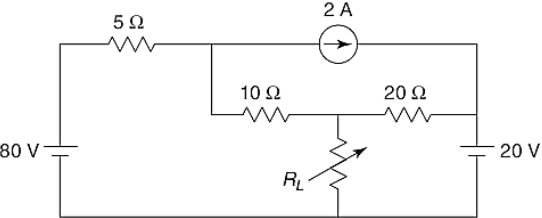
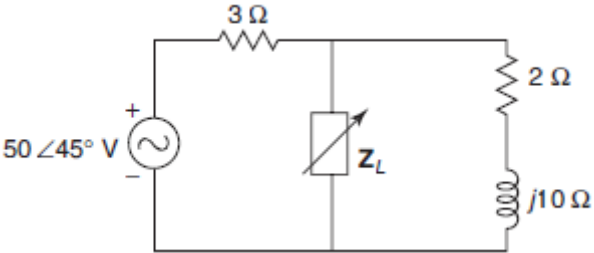
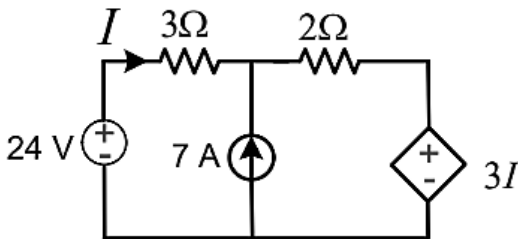
Unit-2

Question No.	Questions	Marks	Course Outcome	RBT Level
1	Find the current through the 10Ω resistor using Norton's theorem. 	6	CO2	L3
2	Find Norton's equivalent network across terminals A and B of the figure 	8	CO2	L3
3	Obtain Norton's equivalent circuit for the network shown 	6	CO2	L3
4	Find the current through $(4+j6)\Omega$ impedance in the network shown using Thevenin's theorem.	6	CO2	L3

				
5	Find the Thevenin and Norton equivalent for the circuit shown in Figure with respect terminals A-B. 	6	CO2	L3
6	Find the current through 5Ω resistor using Norton's theorem for the circuit shown in Figure. 	6	CO2	L3
7	Find the current through 16 Ω resistor using Nortons theorem for the circuit shown in Figure. 	6	CO2	L2
8	State and prove Thevenin's theorem.	6	CO2	L2

Unit-3

Question No.	Questions	Marks	Course Outcome	RB T Level
1	Determine the current through the 10 Ω resistor using superposition theorem.	7	CO2	L3

				
2	Calculate the current I in the network shown in figure using superposition theorem. 	7	CO2	L3
3	State and prove Maximum power transfer theorem When the load impedance is a complex impedance with variable resistance and variable reactance.	7	CO2	L2
4	Find the value of resistance R_L for maximum power transfer and calculate the maximum power. 	10	CO2	L3
5	Find the value of load impedance Z_L so that maximum power can be transferred to it in the network shown. Find maximum power. 	7	CO2	L3
6	Using superposition theorem find the current I in circuit shown in Figure. 	7	CO2	L3
7	Find the node voltage V_2 in the network shown using superposition theorem.	7	CO2	L3

8	State maximum power transfer theorem. For the circuit shown in Figure, what should be the value of R such that maximum power transfer can take place from the rest of the network. Obtain the amount of this power.	7	CO2	L3
9	State and prove Superposition theorem.	7	CO2	L2

MODULE-3: Two port Network Parameters

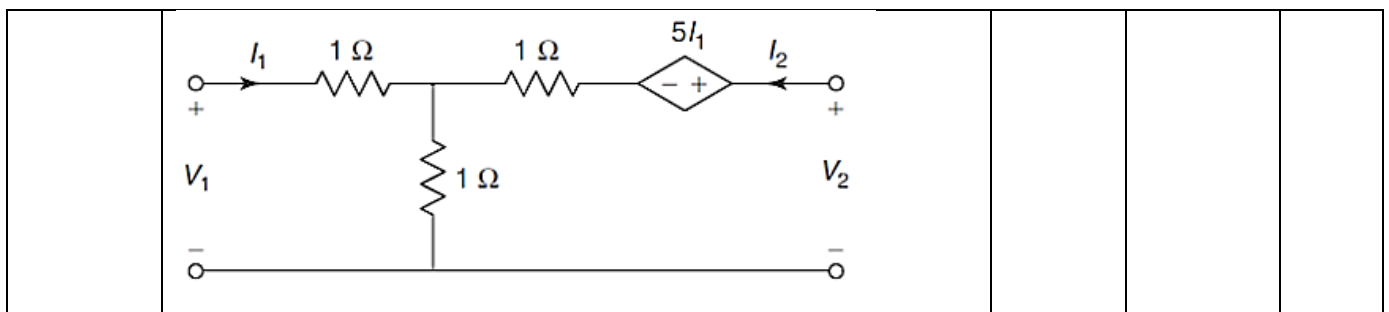
Unit-1

Question No.	Questions	Marks	Course Outcome	RBT Level
1	Express Z- parameters in terms of Y- parameters	7	CO3	L2
2	Express Y- parameters in terms of Z- parameters	7	CO3	L2
3	Express h- parameters in terms of Z- parameters	7	CO3	L2
4	Express T- parameters in terms of Y- parameters	7	CO3	L2
5	Express Y- parameters in terms of h- parameters	7	CO3	L2
6	Express ABCD- parameters in terms of h- parameters	7	CO3	L2
7	Express h- parameters in terms of Y- parameters	7	CO3	L2
8	Express T- parameters in terms of Z- parameters	7	CO3	L2

Unit-2

Question No.	Questions	Marks	Course Outcome	RBT Level
1	Determine Z- Parameters.	7	CO3	L2
2	Find Z parameters for the network shown in figure	7	CO3	L2

3	Determine Y parameter for the network shown in Fig.	7	CO3	L2
4	For the network shown in Fig Find Z and Y parameters	7	CO3	L2
5	Find Y and Z Parameters for the network shown in Figure	7	CO3	L2
6	Find Z parameters for the network shown in Figure and hence find h Parameters	7	CO3	L2
7	Find the Z and Y Parameter for the network shown in Figure	7	CO3	L2
8	Determine Y-Parameters	7	CO3	L2



Unit-3

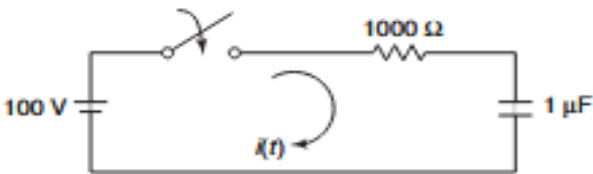
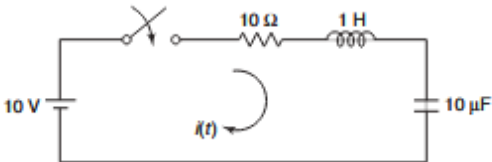
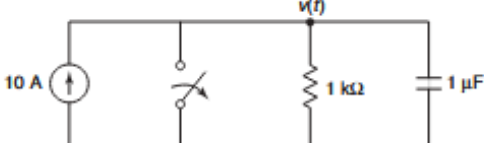
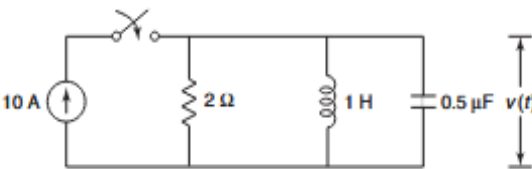
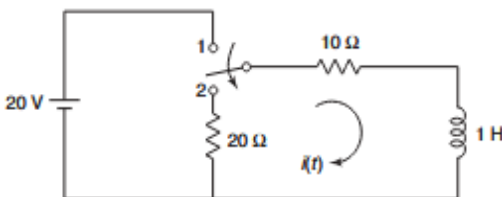
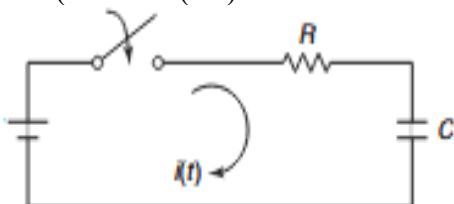
Question No.	Questions	Marks	Course Outcome	RBT Level
1	Find the Transmission parameter for the network shown in Fig	6	CO3	L2
2	Find the Transmission parameter for the network shown in Figure	6	CO3	L2
3	Find h - parameters for the following circuit.	8	CO3	L2
4	Find the Z and Y parameter for the network shown in Figure	8	CO3	L2
5	Determine h-Parameters	6	CO3	L2

6	Find h-Parameters.	6	CO3	L2
7	Following short circuit currents and voltages are obtained experimentally for a two port network: Determine Y parameters i) With output short circuited $I_1 = 5\text{mA}$; $I_2 = -0.3\text{mA}$ and $V_1 = 25\text{ V}$ ii) With input short circuited $I_1 = -5\text{mA}$; $I_2 = 10\text{mA}$ and $V_2 = 30\text{ V}$	6	CO3	L2
8	Following are the hybrid parameters for a network Determine the Y parameters for the network $\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix}$	6	CO3	L2
9	Determine h-Parameters	6	CO3	L2

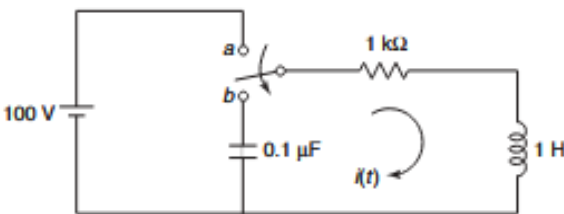
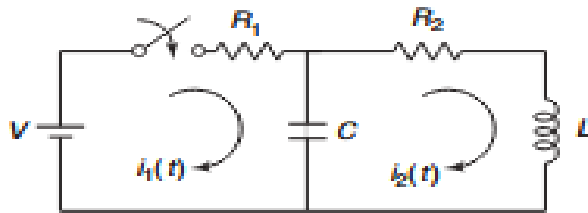
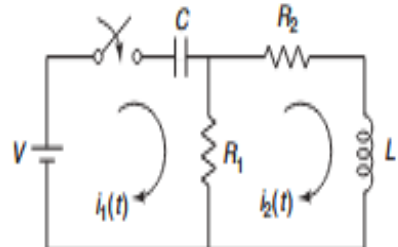
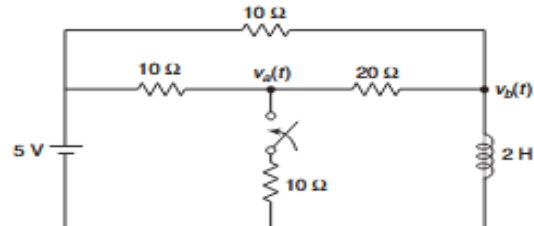
MODULE-4: Transient behaviour and initial conditions

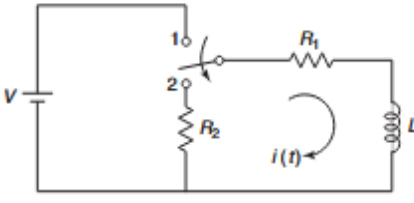
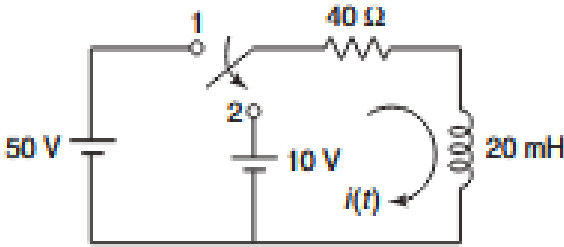
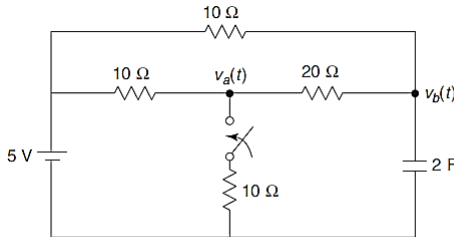
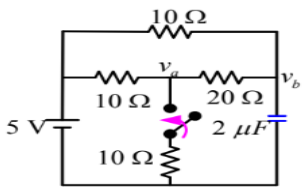
Unit-1

Question No.	Questions	Marks	Course Outcome	RBT Level
1	In the given network shown in figure the switch is closed at $t=0$. With zero current in the inductor, find i, di/dt and d^2i/dt^2 at $t=0^+$	6	CO4	L2
2	For the network shown in figure, the switch is closed at $t=0$. With the capacitor uncharged, find value for I, di/dt and d^2i/dt^2 at $t=0^+$	6	CO4	L2

				
3	In the network shown in figure, the switch is closed. Assuming all initial conditions as zero find I, di/dt and d^2i/dt^2 at $t=0^+$ 	6	CO4	L2
4	In the given network shown in figure, the switch is opened at $t=0$. Solve for V, dv/dt and d^2v/dt^2 at $t=0^+$ 	6	CO4	L2
5	For the network shown in figure, the switch is closed at $t=0$, determine v, dv/dt and d^2v/dt^2 at $t=0^+$ 	6	CO4	L2
6	In the network shown in figure, the switch is changed from position 1 to the position 2 at $t=0$, steady condition having reached before switching. Find the values of i, di/dt and d^2i/dt^2 at $t=0^+$ 	8	CO4	L2
7	Write the behaviour of R, L, C elements at the time of switching, at $t=0$ both at $t=0^+$ and $t=\infty$	7	CO4	L2
8	For the circuit shown below, the switch is closed at $t=0$. Find the initial conditions at $i(0^+)$ and $d(i0^+)/dt$ 	7	CO4	L2

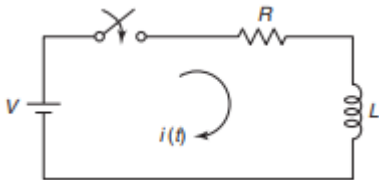
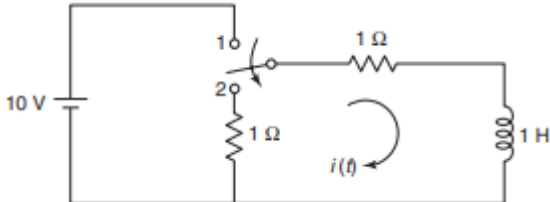
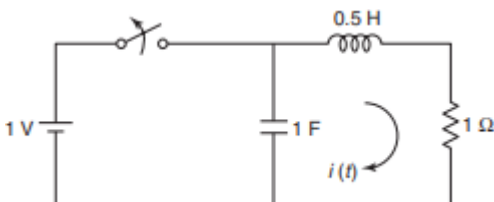
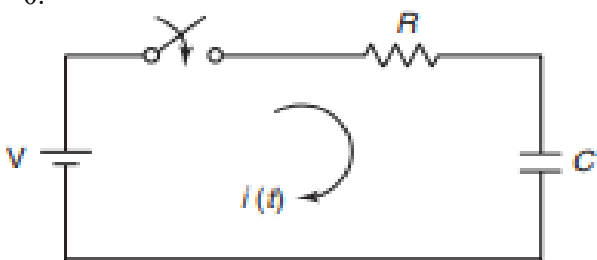
Unit-2

Question No.	Questions	Marks	Course Outcome	RBT Level
1	In the network shown, the switch is changed from the position 'a' to 'b' at $t=0$. Solve for i, di/dt and d^2i/dt^2 at $t=0^+$ 	7	CO4	L2
2	In the network shown in figure. Assuming all initial conditions as zero, find $i_1(0^+)$, $i_2(0^+)$, $\frac{di_1}{dt}(0^+)$, $\frac{di_2}{dt}(0^+)$, $\frac{d^2i_1}{dt^2}(0^+)$ and $\frac{d^2i_2}{dt^2}(0^+)$. 	7	CO4	L2
3	In the network shown in figure assuming all initial conditions as zero find for i_1, i_2, di_1/dt, di_2/dt, d^2i_1/dt^2 and d^2i_2/dt^2 at $t=0^+$ 	7	CO4	L2
4	In the network shown in figure, a steady state is reached with the switch open. At $t=0$, the switch is closed. For the element values given, determine the value of $V_a(0^-)$, $V_b(0^-)$, $V_a(0^+)$ and $V_b(0^+)$ 	7	CO4	L2
5	In the network shown in figure, the switch is initially at the position 1. On the steady state having reached, the switch is changed to the position 2. Find the current $i(t)$	7	CO4	L2

				
6	For the network shown in figure it is under steady state with switch at the position 1. At $t=0$, switch is moved to position 2 find $i(t)$. 	7	CO4	L2
7	In the network shown in figure, a steady state is reached with the switch open. At $t=0$, the switch is closed. For the element values given, determine the value of $V_a(0^-)$, $V_b(0^-)$, $V_a(0^+)$ and $V_b(0^+)$ 	7	CO4	L2
8	In the circuit shown the steady state is reached with switch is open. At $t=0$, switch K is closed. Find the value of $V_a(0^-)$ and $V_a(0^+)$ 	7	CO4	L2

Unit-3

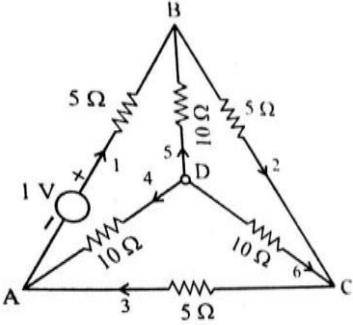
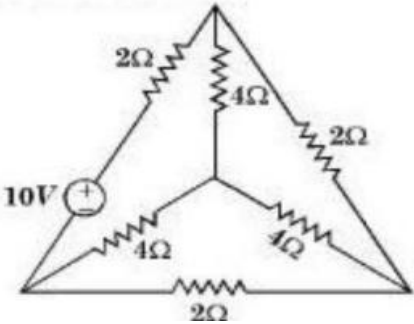
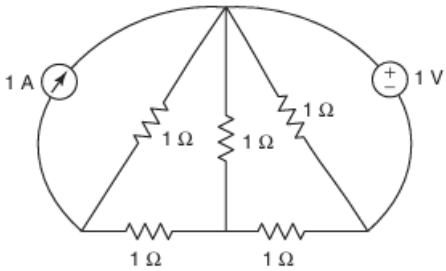
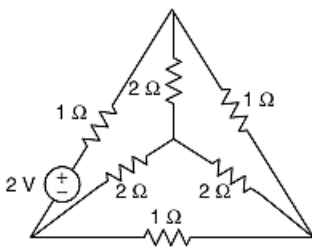
Question No.	Questions	Marks	Course Outcome	RBT Level
1	Find $i(t)$ for series RL circuit as shown in Figure. The switch is closed at time $t = 0$.	7	CO4	L3

				
2	In the network of Fig. 7.12, the switch is moved from the position 1 to 2 at $t = 0$, steady-state condition having been established in the position 1. Determine $i(t)$ for $t > 0$. 	7	CO4	L3
3	In the network shown in Figure, the switch is opened at $t = 0$. Steady-state condition is achieved before $t = 0$. Find $i(t)$. 	7	CO4	L3
4	Find $i(t)$ for series RC circuit as shown in Figure. The switch is closed at time $t = 0$. 	7	CO4	L3

MODULE-5: Graph Theory and Network equations

Unit-1

Question No.	Questions	Marks	Course Outcome	RBT Level
1	Explain the following with respect to Network topology (i) Planar and Non Planar graph (ii) Incidence matrix (iii) Number of possible trees	10	CO5	L2
2	For the network shown in fig, write the tie set schedule and obtain equilibrium equation using KVL. Calculate loop currents. Follow the same orientation and branch numbering as shown in fig. Use branches 4,5,6 as tree branches.	10	CO5	L3

				
3	Draw the oriented graph of a network shown in figure. Select a tree, write the tieset schedule and obtain equilibrium equation. 	10	CO5	L3
4	For the network shown in Fig, draw the oriented graph. Write the tieset schedule and hence obtain the equilibrium equation on loop basis. Calculate the values of branch currents and branch voltages. 	10	CO5	L3
5	For this network shown in Figure, write down the tieset matrix and obtain the network equilibrium equation using KVL. Calculate the loop currents and branch currents. 	10	CO5	L3
6	For the network shown in Figure, write down the tieset matrix and obtain the network equilibrium equation in matrix form using KVL. Calculate loop currents.	10	CO5	L3

7	For the networks shown in Fig, write the incidence matrix, tieset matrix.	5	CO5	L2
8	Find the maximum possible number of trees for the network shown in figure.	5	CO5	L3

Unit-2

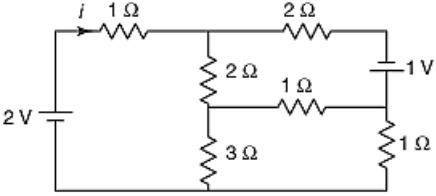
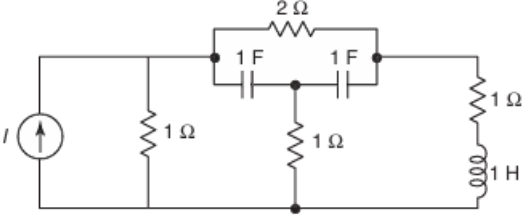
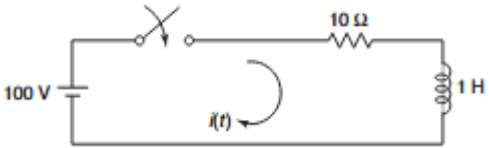
Question No.	Questions	Marks	Course Outcome	RBT Level
1	For the circuit shown in figure, the ohmic values also represent the branch currents. Form a tree using 4Ω, 5Ω and 6Ω branches find the branch currents using cut set matrix.	10	CO5	L3
2	Draw the oriented graph of a network shown in figure. Select a tree, write the cutset schedule and obtain equilibrium equation.	10	CO5	L3

3	For this network shown in Figure, write down the cutset matrix and obtain the network equilibrium equation in matrix form using KCL.	8	CO5	L3																																																															
4	For the network shown in Figure, write down the cutset matrix and obtain the network equilibrium equations.	10	CO5	L3																																																															
5	Explain the concept of tieset schedule and cutset schedule with an example.	10																																																																	
6	Explain the following with respect to Network topology (i) Oriented graph (ii) Tree (iii) Link (iv) Co-tree (v) Connected graph	10																																																																	
7	Draw the oriented graph from the complete incidence matrix given below. <table><tr><th>Nodes ↓</th><th colspan="8">Branches →</th></tr><tr><th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>2</td><td>0</td><td>1</td><td>0</td><td>0</td><td>-1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>3</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>-1</td><td>1</td><td>-1</td></tr><tr><td>4</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>-1</td><td>0</td></tr><tr><td>5</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>	Nodes ↓	Branches →									1	2	3	4	5	6	7	8	1	1	0	0	0	1	0	0	1	2	0	1	0	0	-1	1	0	0	3	0	0	1	0	0	-1	1	-1	4	0	0	0	1	0	0	-1	0	5	-1	-1	-1	-1	0	0	0	0	5	CO5	L2
Nodes ↓	Branches →																																																																		
	1	2	3	4	5	6	7	8																																																											
1	1	0	0	0	1	0	0	1																																																											
2	0	1	0	0	-1	1	0	0																																																											
3	0	0	1	0	0	-1	1	-1																																																											
4	0	0	0	1	0	0	-1	0																																																											
5	-1	-1	-1	-1	0	0	0	0																																																											
8	Determine and draw the number of possible trees for the graph shown.	5	CO5	L2																																																															

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Unit-3

Question No.	Questions	Marks	Course Outcome	RBT Level
1	For the network shown in figure, Draw its dual. Write the integro differential form (i) Mesh equations for given network (ii) Node equations for the dual. $V(t) = 10 \sin 40t$	10	CO5	L3
2	Draw the dual of the network shown in Figure.	5	CO5	L3
3	What are dual networks? What is their significance? Draw the dual of the circuit shown in fig.	5	CO5	L3
4	Draw the dual for the circuit shown in figure	5	CO5	L2

5	Draw the dual for the circuit shown in figure 	5	CO5	L2
6	Draw the dual for the circuit shown in figure 	5	CO5	L2
7	Draw the dual for the circuit shown in figure 	5	CO5	L2
8	With an example explain the concept of Duality.	5	CO5	L2